

Limo Booking Web Application – Project Blueprint (Version 1.0)

Project Overview

Project Goal

Develop a **simple, modern, and budget-friendly limousine booking website** that enables customers to request a ride online while allowing the business owner to manage bookings manually.

The website is **not** intended to be an Uber/Lyft-style application. It will simply automate customer inquiries, reduce phone calls, and notify both the customer and the business owner instantly after a booking request is submitted.

The owner will personally communicate with customers to finalize pricing, availability, and payment.

Business Objectives

- Build a professional online presence
 - Increase online bookings
 - Reduce manual booking effort
 - Automate customer confirmations
 - Improve customer experience
 - Keep monthly operating costs very low
 - Keep the application simple and easy to maintain
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Target Customers

- Airport Travelers
 - Corporate Travelers
 - Families
 - Hotel Guests
 - Wedding Clients
 - Business Executives
 - Tourists
 - Long Distance Travelers
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Technology Stack

Frontend

- Spring Boot Thymeleaf
 - Bootstrap 5
 - HTML5
 - CSS3
 - JavaScript
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Backend

- Java 17
 - Spring Boot 3.x
 - Spring Data JPA
 - Spring Validation
 - Maven
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Database

PostgreSQL

Messaging

Twilio WhatsApp Business API

Email

SMTP (Gmail or Business Email)

Maps

Mapbox

Reason:

- Free tier suitable for small businesses
- Address autocomplete

- Distance calculation
 - Route information
 - Lower cost than Google Maps for small projects
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Hosting

AWS Lightsail / EC2

Application Modules

1. Home Page

Features

- Hero Banner
 - Call Now Button
 - Book Ride Button
 - Featured Services
 - Luxury Fleet Preview
 - Why Choose Us
 - Customer Reviews
 - Google Rating
 - Contact Information
-

2. Fleet Page

Display all available vehicles.

Each vehicle includes:

- Image
- Vehicle Name
- Passenger Capacity
- Luggage Capacity
- Description

Vehicles

- Sedan
- SUV
- Luxury Sedan
- Luxury SUV

3. Services Page

Display available services.

Examples

- Airport Transfer
 - Point to Point
 - Hourly Service
 - Corporate Travel
 - Wedding Transportation
 - Special Events
 - Long Distance
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4. Pricing Page

Display pricing information.

Airport → City

Vehicle	Price
Sedan	\$95
SUV	\$110
Luxury Sedan	\$110
Luxury SUV	\$135

Airport → Suburban Area

Vehicle	Price
Sedan	\$3 / mile
SUV	\$4 / mile
Luxury Sedan	\$4.5 / mile
Luxury SUV	\$6 / mile

Long Distance

Display:

Call for Pricing

Additional Charges

Baby Seat

+\$15

Meet & Greet

Domestic

+\$35

International

+\$45

5. Booking Page

Customer enters:

- Name
 - Phone Number
 - Email
 - Pickup Location
 - Drop Location
 - Pickup Date
 - Pickup Time
 - Vehicle Type
 - Number of Passengers
 - Flight Number (Optional)
 - Baby Seat Required
 - Meet & Greet
 - Special Instructions
 - WhatsApp Consent Checkbox
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6. Contact Page

Includes

- Contact Form
 - Business Phone
 - Business Email
 - Business Address
 - Embedded Google Map
 - Business Hours
 - Social Media Links
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7. About Page

Contains

- Company Introduction
 - Mission
 - Experience
 - Why Choose Us
-

8. FAQ Page

Examples

- How do I book?
 - Can I cancel?
 - What payment methods do you accept?
 - Do you provide child seats?
 - Are airport pickups available?
 - How will I receive confirmation?
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Booking Workflow

Customer opens website



Selects vehicle



Enters pickup & drop locations



Mapbox suggests addresses



Customer selects locations



Distance calculated



Estimated fare displayed (if applicable)



Customer submits booking



Spring Boot validates request



Booking saved into database



Booking Reference Generated



Owner receives WhatsApp notification



Customer receives WhatsApp confirmation



Confirmation Email sent

↓

Owner contacts customer personally

↓

Payment collected

↓

Booking marked as Confirmed

Booking Status

Only four statuses:

- Pending
 - Confirmed
 - Completed
 - Cancelled
-

Booking Reference Format

Example

LIMO-20260816-000123

Pricing Logic

Airport Flat Rates

Use predefined pricing.

Airport to Suburb

Calculate

Distance × Per Mile Rate

Long Distance

Display

"Call for Pricing"

Mapbox Features

- Pickup Address Autocomplete
 - Drop Address Autocomplete
 - Address Validation
 - Distance Calculation
 - Estimated Travel Time
 - Route Display (optional)
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WhatsApp Notifications

Business Owner

Receive

- Customer Name
 - Pickup
 - Drop
 - Date
 - Time
 - Vehicle
 - Passengers
 - Booking Reference
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Customer

Receive

- Thank You Message
 - Booking Reference
 - Booking Summary
 - Business Contact Details
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Email Notifications

Customer receives

- Booking Confirmation
- Booking Reference
- Contact Information

Owner receives

- Booking Details
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Admin Panel

Admin Login

Manage

- Bookings
- Fleet
- Pricing
- Website Content
- Contact Information

Features

- Search Booking
 - Update Status
 - Delete Booking
 - Export CSV
-

Database Tables

Customers

- id
- name
- phone
- email

Bookings

- bookingId
- customerId
- pickup
- drop
- date
- time
- vehicle
- passengers
- status
- estimatedFare
- remarks

Vehicles

- id
- name
- capacity
- luggage
- image
- description

Pricing

- id
- routeType
- vehicle
- rateType
- amount

Features Included at No Extra Cost

Customer Experience

- Responsive Mobile Design
- Click to Call
- Click to Email
- WhatsApp Chat Button
- Booking Reference Number
- FAQ Section
- Customer Testimonials
- Fleet Gallery
- Google Reviews Section
- Embedded Google Map

- Download Contact (vCard)
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Technical

- SEO Friendly URLs
 - Meta Tags
 - Sitemap
 - robots.txt
 - Google Analytics
 - CSV Booking Export
 - Email Notifications
 - Form Validation
 - Input Sanitization
 - Error Handling
 - Logging
-

Payment Strategy

Version 1 will **not** include an integrated payment gateway.

Instead:

Customer submits booking

↓

Owner reviews booking

↓

Owner contacts customer

↓

Price confirmed

↓

Owner sends Payment Link

↓

Customer pays



Booking confirmed

Recommended payment providers

- Stripe Payment Links
- Square Payment Links
- PayPal Payment Links

Advantages

- No complex payment integration
- No monthly subscription
- Lower development effort
- Owner maintains pricing control
- Suitable for custom quotes

Security

- HTTPS
- Spring Validation
- CSRF Protection
- Secure Environment Variables
- Input Validation
- SQL Injection Protection
- XSS Protection

Future Enhancements (Version 2)

- Customer Login
 - Online Payments
 - Promo Codes
 - Loyalty Program
 - Booking History
 - Invoice Download
 - Driver Module
 - AI Chatbot
 - Multi-language Support
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Estimated Monthly Cost

Service	Estimated Cost
Domain	~\$10-20/year
Hosting (AWS Lightsail/EC2)	~\$10-25/month
PostgreSQL	Free (self-hosted) or low-cost managed
Twilio Phone Number	~\$1-2/month
WhatsApp Messages	Based on usage
Mapbox	Free tier initially
SMTP Email	Free
SSL Certificate	Free (Let's Encrypt)
Google Analytics	Free

Expected operating cost for a new business:

Approximately \$20-50/month, excluding WhatsApp message charges and payment gateway transaction fees.

Development Plan

Phase 1

- Project Setup
- UI Design
- Website Pages
- Responsive Layout

Phase 2

- Booking Form
 - Validation
 - PostgreSQL Integration
 - Booking APIs
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Phase 3

- Mapbox Integration
 - Fare Estimation
 - Pricing Engine
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Phase 4

- WhatsApp Integration
 - Email Notifications
 - Booking Reference Generation
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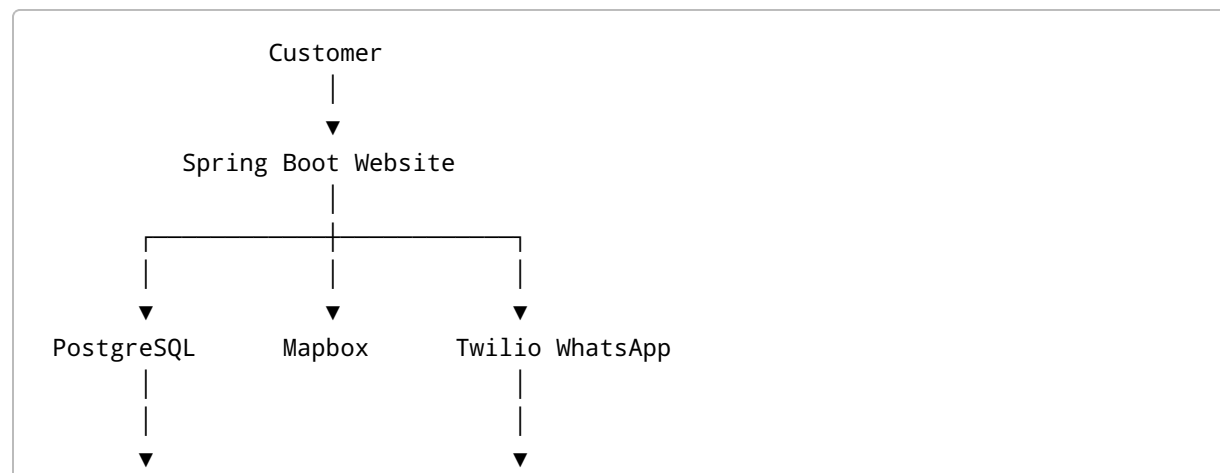
Phase 5

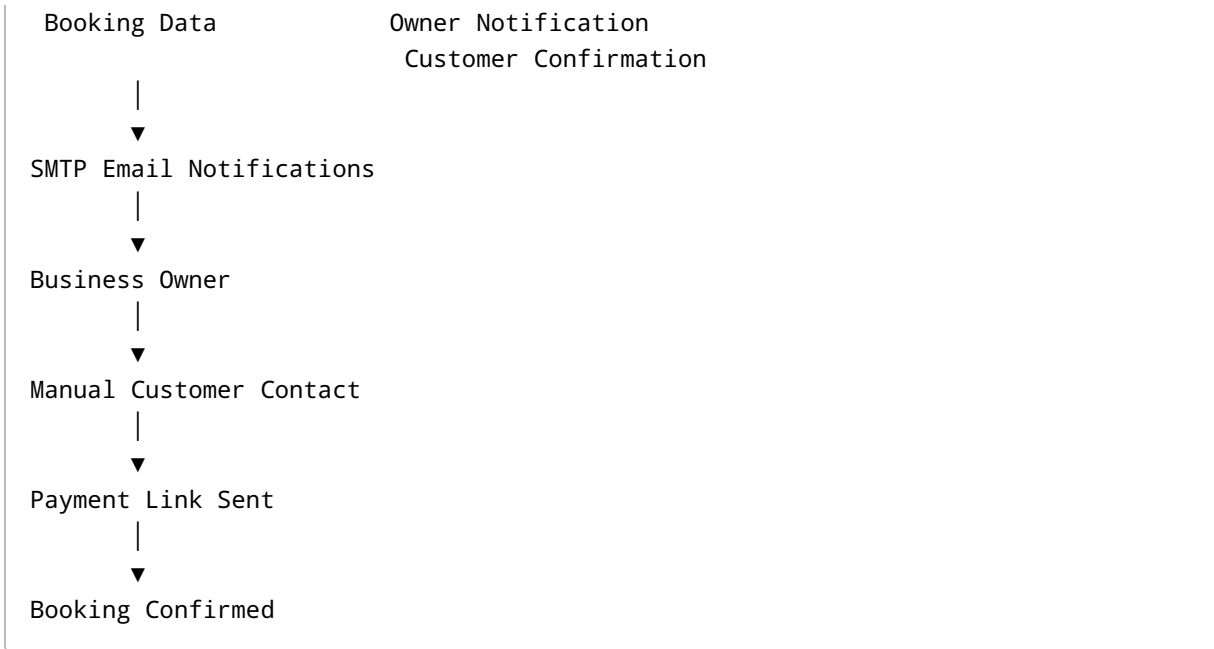
- Admin Panel
 - CSV Export
 - Pricing Management
 - Fleet Management
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Phase 6

- SEO
 - Testing
 - Production Deployment
 - Domain Configuration
 - SSL Setup
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Final Architecture





Success Criteria

The project will be considered successful if it enables customers to easily request a limousine service online, automatically notifies both the customer and business owner, provides a professional online presence, and allows the owner to manage bookings efficiently with minimal operational cost while remaining scalable for future enhancements.